

GOVERNMENT ENGINEERING COLLEGE, SHEIKHPURA

B. Tech 2nd Semester Mid-Sem Exam, 2026

Branch: CSEA, ECE & EE

Sem: 2nd

Time: 2hrs

Full Marks: 20

Sub: Mathematics-II (Complex Analysis & Ordinary Differential Equation)

Instructions:

- i. The marks are indicated in the right-side margin.
- ii. There are 6 questions in this question paper.
- iii. Answer any 4 questions.
- iv. Question no.1 is compulsory.

[SECTION-A]

1. Choose the correct answer of the following (Answer any five).

- (i) The principal value of $\text{Log}(-e^2)$
(a) 2 (b) $2 + \pi i$ (c) $2 - \pi i$ (d) -2 CO1, E [1 mark]
- (ii) If $u(x, y) = 2x - x^2 + my^2$ is Harmonic, then m is
(a) -1 (b) 1 (c) 2 (d) 0 CO1, AP [1 mark]
- (iii) If $f(z) = x^2 - y^2 + x + i(2xy + y)$ is differentiable, then $f'(-i)$ is
(a) $2i$ (b) i (c) $-2i$ (d) $-i$ CO1, AP [1mark]
- (iv) The function $f(z) = |z|^2$ is
(a) everywhere analytic (b) nowhere analytic
(c) analytic at $z = 0$ (d) None of these CO1, U [1mark]
- (v) The value of $\oint_C \frac{1}{z+1} dz$, where $C: |z| = 2$, is
(a) 0 (b) $2\pi i$ (c) πi (d) None of these CO2, AP [1 mark]
- (vi) Consider the function $f(z) = \frac{z}{z^2 \sin(z)}$. Then $z = 0$ is
(a) a pole of order 1 (b) a pole of order 2
(c) Essential singularity (d) Removable Singularity CO2, U [1mark]
- (vii) If $f(z)$ is an analytic at $z = 1$, then $\text{Res}(f, 1)$ is
(a) 1 (b) 0 (c) ∞ (d) NOT CO2, R [1 mark]

[SECTION-B]

Note: Answer any three questions.

2. Show that $u(x, y) = \frac{y}{x^2+y^2}$ is Harmonic and find its Harmonic conjugate $v(x, y)$.

Also, determine the analytic function $f = u + iv$.

CO1, C [5 marks]

3. Find residues at each pole of

$$f(z) = \frac{5z^2 - 4z + 3}{(z+1)(z+2)(z+3)}$$

CO2, AP [5 marks]

4. Find the Laurent series expansion of $f(z) = \frac{1}{z(z-3)}$, valid in the region

$$1 < |z + 1| < 4.$$

CO2, U [5 marks]

5. Evaluate the integral

$$\oint_C \frac{z}{z^4 - 1} dz$$

where $C: |z| = 2$.

CO2, E [5 marks]

6. Find residues at each pole of

$$f(z) = \frac{1}{z \sin(z)}.$$

CO2, U [5 marks]

*****Best of Luck*****